

Active and reactive cell shape change in silico

S. Utecht^{a,b,*} and J. Schauser^a

^a *University of Copenhagen (UCPH), Copenhagen, Denmark*

^b *University of Copenhagen (UCPH), Copenhagen, Denmark*

1. Introduction

hero

- Morphogenesis & tissue adaptability
- Active and adaptive cells
- The polar-adhesion model
- Other modelling approaches - Where is the field now?

2. Active cell-shape change

The polarity adhesion model as first presented in [REFS] does ..

*Corresponding author. E-mail address: sebastian.utecht@ku.nbi.dk

2.1. Cell curvature

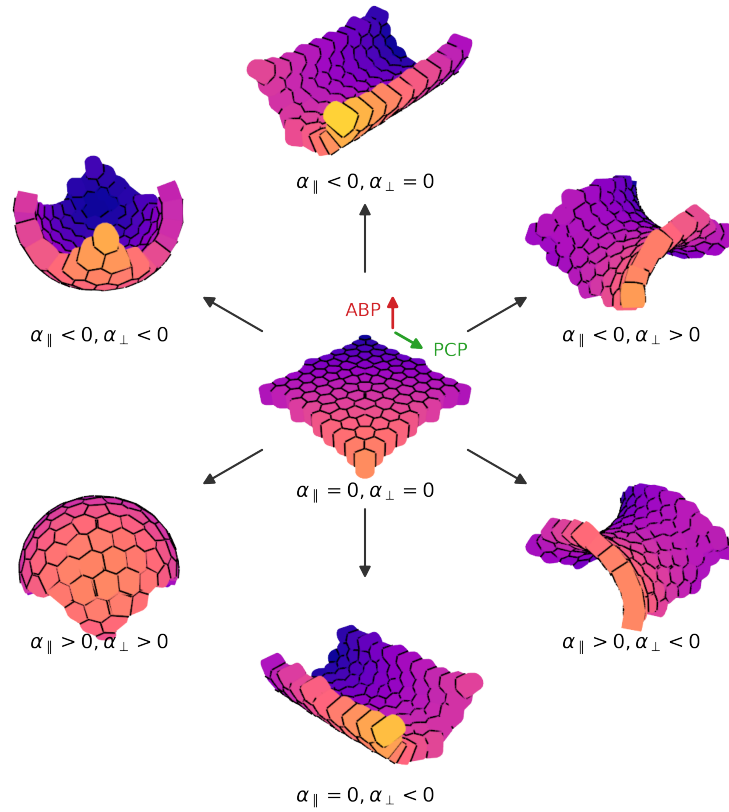


Figure 1: Resulting curvature in a ten by ten sheet of cells for different values of $\alpha_{\parallel}$ and $\alpha_{\perp}$ [PROBABLY MORE]. Coloring is displaying depth.

2.2. *Cell elongation*

- Figure 2: Result of 10x10 sheet under different lambda values + cartoon of cell elongation.

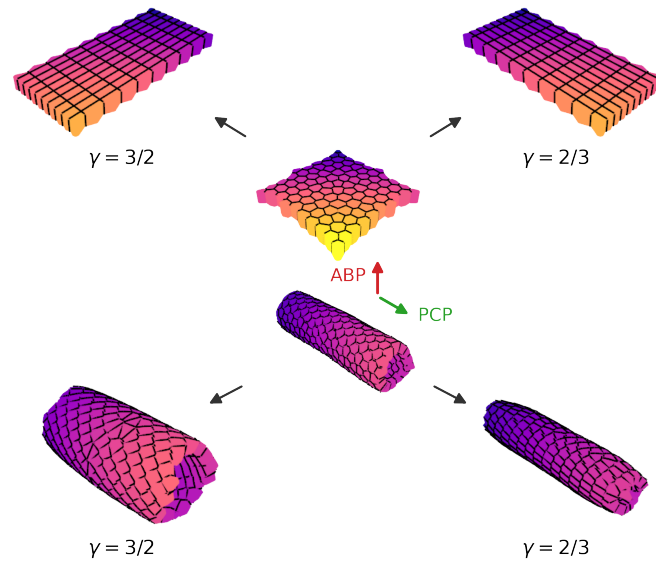


Figure 2: Resulting elongations of simulated cell tissue in a ten by ten sheet of cells and 15 by 20 cylinder for different values of γ . For $\gamma < 1$ the cells are contracting along PCP direction and expanding perpendicular to it, for $\gamma > 1$ the opposite is true [PROBABLY MORE]. Coloring is displaying depth.

3. Reactive shape change

3.1. Reactive curvature

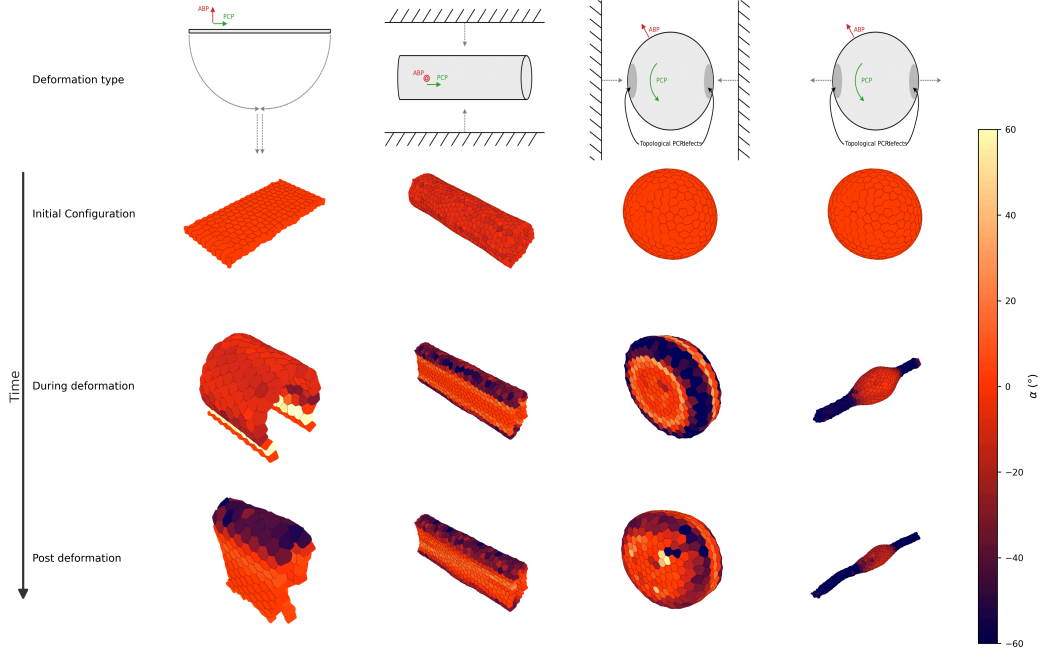


Figure 3: Figure displays the effects of various mechanic deformations on various simulated cellular structures when $\alpha_{\parallel}$ and $\alpha_{\perp}$ are ‘free’ parameters, i.e. eq [REF] is included in the update equations. Coloring displays the value of either $\alpha_{\parallel}$ or $\alpha_{\perp}$ in degrees (see colorbar) depending on which is most important for the deformation shown. *Column 1*: A sheet of 20x10 cells is deformed by dragging the edge of the shortside of the sheet in a halfcircle and then down. The cells near the fold adapts and display much higher $\alpha_{\parallel}$ values than the remaining sheet. The resulting structure is stable. *Column 2*: A cylinder is squeezed between two infinite planes. $\alpha_{\perp}$ values near the folds are markedly larger and the structure is stable. *Column 3*: A sphere is squeezed between two infinite planes, again the cells around the folds adapt, but the structure is not completely stable and ‘bounces back’ a bit to become stable. *Column 4*: A sphere is stretched by dragging on a subset of cells on each side of it. The structure becomes increasingly tube-like as stretched and is stable.

3.2. Reactive elongation

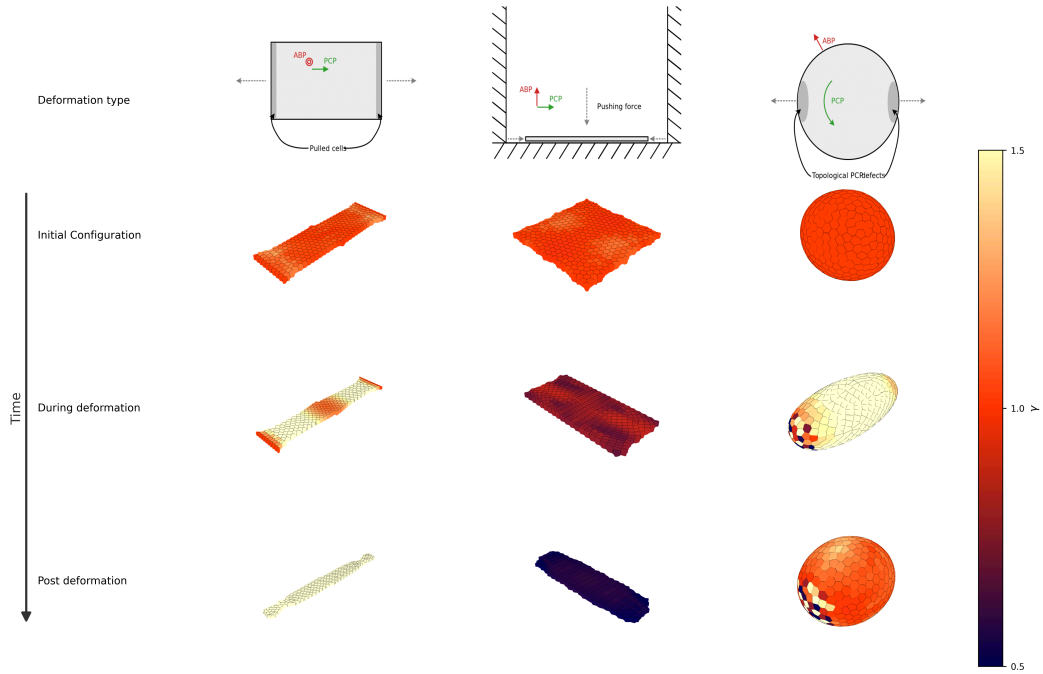


Figure 4: Figure displays the effects of various mechanic deformations on simulated cellular structures when γ is a ‘free’ parameters, i.e. eq [REF] is included in the update equations. Coloring of cells displays values of γ . *Column 1:* A 20x10 sheet of cells is pulled. [MORE]

3.3. Combining curvature and elongation

- Figure 5: Ball being pulled with learnable alphas and gammas

4. Active and adaptive domains

4.1. The case of neurolation

Doth this work? How does it work? What are the implications for the field?

4.2. *The case of something else*

5. Discussion

6. Conclusion

Heyo what is up baybee

7. Changes needed to be made

- **Figures:**
 - Consider keeping ABP and PCP as unit vectors on sphere figure
 - Include scale bars in all illustrations and seek to keep them as similar as possible
 - Keep all variables as close to each other as possible. Might need to redo some simulations here due to noise, deformation times, cell counts etc. Look into this.
 - Mark subset of pulled cells AND subset of screened out defect cells where applicable
 - Make null-case figures for everything
 - For figure 4: Redo stretch straight stretch such that the initial configuration doesn't look wonky
 - For figure 3+4: Add additional row displaying the peak deformation before it is let go.
 - Figure 1 should be reduced to 3 curvatures (as all others are rotations) and we need to add illustrations of cells wedging along the arrows) We also need to add cells elongating along the arrows of figure 2
- **Results:**
 - Write out the math and be sure of it.
 -

References